

VLSI Template Stage Output Matrix

Board-specific VLSI flow view for the July_04 engineering review package.

Template Stage	Board-Specific Output	Files	Status	Engineer Action
1. Front-End Design	Schematic closure package	04_front_end_schematic, 07_netlists, 16_verification	214 schematic-parity warnings remain.	Provide complete schematic/netlist source and close or sign-waive parity.
2. Back-End Design	PCB ECO package	08_board_outline_mechanical, 10_component_placement, 11_routing_planes	0 DRC errors; 82 library/footprint mismatch warnings.	Resolve library mappings and confirm chassis/mechanical constraints.
3. Analog Design	TFLN optical/RF/bias review	04_front_end_schematic, 06_footprints_symbols, 12_silkscreen_drc_fab_notes	Review output issued.	Validate TFLN PIC, RF drivers, bias rails, monitor lines, and MZM lanes.
4. Mixed-Signal Design	NCE-to-TFLN interface review	04_front_end_schematic, 07_netlists	Interface mapping requires authoritative netlist.	Confirm TFLN_TX lane mapping plus SPI/I2C/JTAG/PLL/AST2600 interfaces.
5. Verification Domain	Validation reports and issue logs	16_verification	0 unconnected items; parity still open.	Target parity zero using complete schematic/netlist source.
6. Physical Verification	DRC, board outline, mechanical, fab notes	08_board_outline_mechanical, 12_silkscreen_drc_fab_notes, 16_verification	Current DRC has 82 warnings.	Rerun KiCad DRC after footprint/library and mechanical closure.
7. FPGA Design	Interface reserve	02_vlsi_domain_outputs/FPGA_DOMAIN_NO TE	No FPGA-specific RTL supplied.	If programmable logic is added, issue RTL, constraints, bitstream, and verification logs.
8. Embedded Systems	AST2600/service-interface checklist	02_vlsi_domain_outputs/EMBEDDED_SYST EMS_SERVICE_CHECKLIST	Checklist output issued.	Confirm firmware ownership, boot straps, management bus, sensors, reset, and update path.
9. CAD / EDA Tool Development	KiCad project, generated reports, manifest	00_OPEN_IN_KICAD, FULL_PACKAGE_MANIFEST	KiCad-compatible handoff format.	Use KiCad 10.0.3 or later and preserve checksum manifest.
10. Semiconductor Fabrication	PCB fab/assembly review package	13_gerbbers_drill, 14_dfa_assembly, 15_dfm_manufacturability, 17_release_gate	DFM/DFA review package issued.	Complete DFM/DFA, BOM owner signoff, and parity closure before production package approval.